

Multivariable Calculus Derivation

Gradient of Categorical Cross-Entropy with Softmax Activation

Z7-OS Machine Learning Research

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The Goal

Prove rigorously using multivariable calculus that for a single sample:

$$\frac{\partial L}{\partial z_i} = a_i - y_i \iff \frac{\partial L}{\partial Z} = a - y$$

1 Definitions and Mathematical Setup

Let:

- $Z = [z_1, z_2, \dots, z_K]^T \in \mathbb{R}^K$ be the pre-activation logit vector for K classes.
- $a = [a_1, a_2, \dots, a_K]^T \in \mathbb{R}^K$ be the Softmax probability vector:

$$a_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} = \frac{e^{z_i}}{\Sigma_Z}, \quad \text{where } \Sigma_Z = \sum_{j=1}^K e^{z_j}$$

- $y = [y_1, y_2, \dots, y_K]^T$ be the true one-hot target vector, satisfying:

$$\sum_{k=1}^K y_k = 1 \quad \text{and} \quad y_k \in \{0, 1\}$$

- The Categorical Cross-Entropy Loss L is defined as:

$$L(a, y) = - \sum_{k=1}^K y_k \ln(a_k)$$

2 Step 1: The Jacobian Matrix of Softmax $\left(\frac{\partial a_k}{\partial z_i} \right)$

Notice that unlike element-wise activations (e.g., Sigmoid, ReLU), Softmax couples all outputs through the shared normalization denominator Σ_Z . Changing logit z_i alters **every single** output probability a_k .

We must analyze two distinct cases using the quotient rule:

$$\frac{\partial}{\partial z_i} \left[\frac{u}{v} \right] = \frac{u'v - uv'}{v^2}, \quad \text{with } u = e^{z_k}, \quad v = \Sigma_Z$$

Case A: When $k = i$ (Diagonal elements of Jacobian)

Here $u = e^{z_i}$, so $\frac{\partial u}{\partial z_i} = e^{z_i}$. Also $\frac{\partial \Sigma_Z}{\partial z_i} = e^{z_i}$.

$$\begin{aligned}\frac{\partial a_i}{\partial z_i} &= \frac{e^{z_i} \Sigma_Z - e^{z_i} e^{z_i}}{\Sigma_Z^2} \\ &= \frac{e^{z_i}}{\Sigma_Z} \left(\frac{\Sigma_Z - e^{z_i}}{\Sigma_Z} \right) \\ &= a_i(1 - a_i)\end{aligned}$$

Case B: When $k \neq i$ (Off-diagonal cross terms of Jacobian)

Here $u = e^{z_k}$ is independent of z_i , so $\frac{\partial u}{\partial z_i} = 0$. However, $v = \Sigma_Z$ still contains e^{z_i} , so $\frac{\partial \Sigma_Z}{\partial z_i} = e^{z_i}$.

$$\begin{aligned}\frac{\partial a_k}{\partial z_i} &= \frac{0 \cdot \Sigma_Z - e^{z_k} e^{z_i}}{\Sigma_Z^2} \\ &= - \left(\frac{e^{z_k}}{\Sigma_Z} \right) \left(\frac{e^{z_i}}{\Sigma_Z} \right) \\ &= -a_k a_i\end{aligned}$$

3 Step 2: Derivative of Loss with Respect to Activations $\left(\frac{\partial L}{\partial a_k} \right)$

Differentiating the cross-entropy loss with respect to a single activation a_k :

$$\frac{\partial L}{\partial a_k} = \frac{\partial}{\partial a_k} [-y_k \ln(a_k)] = -\frac{y_k}{a_k}$$

4 Step 3: Multivariable Chain Rule for $\frac{\partial L}{\partial z_i}$

Because z_i influences every output $a_1, a_2, \dots, a_K$, the multivariable chain rule requires summing over all output nodes k :

$$\frac{\partial L}{\partial z_i} = \sum_{k=1}^K \frac{\partial L}{\partial a_k} \frac{\partial a_k}{\partial z_i}$$

Split the sum into the $k = i$ term and all other $k \neq i$ terms:

$$\frac{\partial L}{\partial z_i} = \left(\frac{\partial L}{\partial a_i} \frac{\partial a_i}{\partial z_i} \right) + \sum_{k \neq i} \left(\frac{\partial L}{\partial a_k} \frac{\partial a_k}{\partial z_i} \right)$$

Substitute the derivatives obtained in Steps 1 and 2:

$$\frac{\partial L}{\partial z_i} = \left(-\frac{y_i}{a_i} \right) \cdot [a_i(1 - a_i)] + \sum_{k \neq i} \left(-\frac{y_k}{a_k} \right) \cdot [-a_k a_i]$$

Simplify both terms algebraically:

$$\begin{aligned}\frac{\partial L}{\partial z_i} &= -y_i(1 - a_i) + \sum_{k \neq i} y_k a_i \\ &= -y_i + y_i a_i + a_i \sum_{k \neq i} y_k \\ &= -y_i + a_i \left(y_i + \sum_{k \neq i} y_k \right)\end{aligned}$$

Notice the expression inside the parentheses:

$$y_i + \sum_{k \neq i} y_k = \sum_{k=1}^K y_k$$

Since y is a one-hot distribution, the sum of all its elements is exactly 1:

$$\sum_{k=1}^K y_k = 1$$

Therefore:

$$\begin{aligned} \frac{\partial L}{\partial z_i} &= -y_i + a_i(1) \\ \frac{\partial L}{\partial z_i} &= a_i - y_i \quad \blacksquare \end{aligned}$$

5 Step 4: Vectorized Form for Batches

In compact vector notation for a single sample:

$$\frac{\partial L}{\partial Z} = a - y$$

For a mini-batch of m samples, the total cost is $J = \frac{1}{m} \sum_{s=1}^m L^{(s)}$. By linearity:

$$\frac{\partial J}{\partial Z} = \frac{1}{m}(A - Y) \in \mathbb{R}^{m \times K}$$

where $A \in \mathbb{R}^{m \times K}$ are the Softmax predictions and $Y \in \mathbb{R}^{m \times K}$ is the one-hot target matrix.